

Arjuna JEE 2025

Mathematics

DPP: 4

Permutations and Combinations

Q1 The number of ways in which ten candidates $A_1, A_2, \dots, A_{10}$ can be ranked such that A_1 is always above A_{10} is

- (A) $5!$
 (B) $2(5!)$
 (C) $10!$
 (D) $\frac{1}{2}(10!)$

Q2 Answer the following questions:

- (i) How many different words can be formed with the letters of the word HARYANA?
 (ii) How many of these being with H and end with N?
 (iii) In how many of these H and N are together?

Q3 The number of permutations of the word 'AUROBIND' in which vowels appear in an alphabetical order is $\frac{8!}{k!}$, then k is

Q4 Six identical coins are arranged in a row. The total number of ways in which the number of heads is equal to the number of tails, is

- (A) 9
 (B) 20
 (C) 40
 (D) 120

Q5 The number of arrangements that can be made with the letters of the word 'MATHEMATICS' in which all vowels come together, is

- (A) $\frac{8! \times 4!}{2! \times 2!}$
 (B) $\frac{8! \times 4!}{2! 2! 2!}$
 (C) $\frac{8!}{2! 2! 2!}$
 (D) $\frac{8!}{4! 2! 2!}$

Q6

Number of ways in which 15 different books can be arranged on a shelf so that two particular books shall not be together is

- (A) $14 \times 15!$
 (B) $13 \times 14!$
 (C) $14! \times 15!$
 (D) $(15!)^2$

Q7 If ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800$, then r is

- (A) 39
 (B) 41
 (C) 28
 (D) 43

Q8 Find the number of 7-digit numbers possible whose sum of digits is 61.

- (A) 12
 (B) 24
 (C) 28
 (D) 36

Q9 In how many ways can 5 boys and 3 girls sit in a row so that no two girls are together

- (A) $5! \times 3!$
 (B) ${}^4P_3 \times 5!$
 (C) ${}^6P_3 \times 5!$
 (D) ${}^5P_3 \times 3!$

Q10 How many words can be made from the letters of the word DELHI, if "L" comes in the middle in every word

- (A) 12
 (B) 24
 (C) 60
 (D) 6

Q11 The number of ways in which four S come consecutively in the word MISSISSIPPI, is

- (A) 420
 (B) 840
 (C) 210
 (D) 630



Q12 If a denotes the number of permutations of $x + 2$ things taken all at a time, b the number of permutations of x things taken 11 at a time and c the number of permutations of $(x - 11)$ things taken all at a time such that $a = 182bc$, then value of x is

- (A) 10 (B) 12
(C) 15 (D) 18

Q13 The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN, is

- (A) 360 (B) 192
(C) 96 (D) 48

Q14 If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in dictionary, then the position of the word SMALL, is

- (A) 46th
(B) 56th
(C) 52th
(D) 58th

Q15 Find the rank of the word 'MOTHER' in dictionary format.

- (A) 307 (B) 308
(C) 309 (D) 310

Q16 If the letters of the CORONA are arranged in all possible ways and these words are written in order of a dictionary, then the word CORONA appears at the serial number.

- (A) 108
(B) 110
(C) 106
(D) 112

Q17

There are 5 gentlemen and 4 ladies to dine at a round table. The number of ways in which they can seat themselves so that no two ladies are together, is

- (A) 2880 (B) 576
(C) 1440 (D) 480

Q18 The number of ways in which seven persons can sit around a table so that all shall not have the same neighbors in any two arrangements, is

- (A) 720 (B) 360
(C) 1440 (D) None of these



Answer Key

Q1 (D)

Q2 (i) 840
(ii) 20
(iii) 240

Q3 4

Q4 (B)

Q5 (B)

Q6 (B)

Q7 (B)

Q8 (C)

Q9 (C)

Q10 (B)

Q11 (B)

Q12 (B)

Q13 (C)

Q14 (D)

Q15 (C)

Q16 (A)

Q17 (A)

Q18 (B)



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